Result

The theoretical in-degree probability distribution for node type $x \in \{a, b\}$ in the limit $\lim_{t\to\infty} p(k,t)$ for the heterophilic Price model is:

$$p_{\infty,x}(k) = \begin{cases} p_{0,x} & k = 0 \\ p_{0,x} \cdot \dfrac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)} & k > 0 \end{cases} \tag{1}$$

where $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ is the beta function. We have assumed analytic continuation extending from $k \in \mathbb{N}$ to $k \in \mathbb{R}$.

Parameters

The distribution parameters are:

$$\alpha_x = \frac{\mu_x}{\lambda_x} \tag{2}$$

$$\gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x} \tag{3}$$

$$b_0 = \frac{1}{1 + \mu_x \tilde{Z}} \tag{4}$$

where:

- μ_x is the initial attractiveness parameter for type x nodes
- λ_x is the coupling strength for type x nodes
- $\lambda_x = f_x h_x + (1 - f_x)(1 - h_x)$
- f_x is the fraction of nodes that are of type "x"
- h_x is the probability of a connection between nodes of the same type. We model, for simplicity, the probability of a connection between different type nodes is $1 - h_x$
- $\tilde{Z} = \frac{m}{Z}$ where $Z = g_a\lambda_a + g_b\lambda_b + f_a\mu_a + f_b\mu_b$ is the partition function
- $Z(t) = Z \cdot t$ is the time-dependent partition function
- m is the number of edges added per timestep
- g_x is the asymptotic mean in-degree for type x nodes

Derivation

Suppose the fraction of male AI researchers is $f \in [0, 1]$. There is an affect in the social sciences called homophily...

A graph, the mathematical object whose real-world instantiation is called a network, $G = (V, E)$, consists of a node set V and an edge set E. We want to study a type of graph called a directed graph. In directed graphs, edges are ordered pairs; $(u, v) \neq (v, u)$ of nodes, $u, v \in V$.

Consider a network containing N nodes. This system can be thought of as containing two types of node: an x type node, and nodes that are not type x. It follows therefore,

$$N = N_x + N_{x'} \tag{5}$$

where N_x is the number of type x nodes and $N_{x'}$ is the number of nodes that are not type x.

We can connect these nodes by drawing arrows from a given source node to a given target node. We call these arrows directed edges. The in-degree of a given node is the total number of arrows pointing toward this node, $k^{(\text{in})}$. The out-degree of the node, $k^{(\text{out})}$, is the number of outward arrows originating from the node.

Let $n_x(k, t)$ be the number of x type nodes with an in-degree k at the time t. The expected number of x type nodes with an in-degree k at time $t + 1$ can be expressed as:

$$n_x(k, t+1) = n_x(k, t) + m\Pi_x(k-1, t)n_x(k-1, t) - m\Pi_x(k, t)n_x(k, t) + \delta_{k,0} \tag{6}$$

The term

$$m\Pi_x(k-1, t)n_x(k-1, t) \tag{7}$$

is the expected number of nodes that transition from degree $k - 1$ to degree k in this timestep. Since there are m new edges added by the new node created in the timestep, there are m opportunities for a given x type node to receive a new in-edge. $\Pi_x(k-1, t)$ is the probability that a type x node with degree $k-1$ receives a new in-edge. We omit consideration of the same node gaining two or more in-degrees in a single timestep here since these so-called multiedge terms are an $O(m\Pi)$ correction, which in the long time limit, is negligible [1]. The quantity $n_x(k - 1, t)$ is the number of these x type nodes with in-degree $k - 1$.

It therefore follows, by an identical line of reasoning, that the term,

$$m\Pi_x(k, t)n_x(k, t) \tag{8}$$

represents the expected number of nodes leaving degree k by gaining an edge and moving to degree $k + 1$.

Finally, $\delta_{k,0}$ is a Kronecker delta that accounts for the case that, given the newly generated node is x type, it will be an addition to the $n(k = 0, t)$ count, as the new node will necessarily have an in-degree of 0.

Now, the $\Pi_x(k, t)$ term itself can be expressed as,

$$\Pi_x(k,t) = \frac{f_x h_x k + (1 - f_x)(1 - h_x)k + \mu_x}{Z(t)} \tag{9}$$

where the two terms in the numerator are the respective expected values of the given x node gaining a new in-degree for each case that the new node is also of type x node or not. It is useful to recognise that the numerator can be rewritten as,

$$\Pi_x(k,t) = \frac{\lambda_x k + \mu_x}{Z(t)} \tag{10}$$

where we have factorised the k in-degree terms and labeled the resulting factor λ_x ($\lambda_x \in \mathbb{R}^+$). The $Z(t)$ denominator is the normalization factor. It ensures that the probabilities of all existing nodes being targeted by the new directed edge sum to 1.

$$\sum_{i \in V_x}^{N_x} \Pi_{x,i} + \sum_{i \in V_{x'}}^{N_{x'}} \Pi_{x',i} = 1 \tag{11}$$

which can be written as,

$$\sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} + \sum_{i \in V_{x'}}^{N_{x'}} \frac{\lambda_{x'} k + \mu_{x'}}{Z(t)} = 1 \tag{12}$$

Using this constraint, we can begin to solve for $Z(t)$. If we work under the conjecture that as $t \to \infty$ both the proportion of in-edges and fraction of node types in the model for a given node type tend to constant values,

$$\lim_{t \to \infty} E_x(t) = \langle g_x \rangle t \tag{13}$$

where $\langle g_x \rangle, \langle g_{x'} \rangle \in \mathbb{R}^+$. and,

$$\lim_{t \to \infty} N_x = f_x t \tag{14}$$

We can now rearrange (11) for Z(t). To do this, let us focus on the

$$\sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} \tag{15}$$

term. We can rewrite this as,

$$\sum_{i\in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} = \frac{\lambda_x E_x^{(\mathrm{in})} + \mu_x N_x}{Z(t)} \tag{16}$$

so, when noticing that,

$$\sum_{i\in V_x}^{N_x} k = E_x^{(\mathrm{in})} = \lambda_x \langle g_x \rangle \tag{17}$$

and using an identical argument for the x' nodes, we now have,

$$\frac{(\lambda_x\langle g_x\rangle + \mu_x f_x)t}{Z(t)} + \frac{(\lambda_{x'}\langle g_{x'}\rangle + \mu_{x'} f_{x'})t}{Z(t)} = 1 \tag{18}$$

$$\implies Z(t) = [(\lambda_x\langle g_x\rangle + \lambda_{x'}\langle g_{x'}\rangle) + (\mu_x f_x + \mu_{x'} f_{x'})]t \tag{19}$$

Purely for the sake of neat algebra, it is now useful to motivate,

$$\tilde{Z} = \frac{m}{(\lambda_x\langle g_x\rangle + \lambda_{x'}\langle g_{x'}\rangle) + (\mu_x f_x + \mu_{x'} f_{x'})} \tag{20}$$

So, if we substitute our newly found values into (6), we get,

$$\begin{aligned} n_x(k,t+1) = n_x(k,t) &+ \tilde{Z}t^{-1}(\lambda_x(k-1)+\mu_x)n_x(k-1,t) \\ &- \tilde{Z}t^{-1}(\lambda_x(k)+\mu_x)n_x(k,t) \end{aligned} \tag{21}$$

and since this is the expected number of nodes with an in-degree k at time $t+1$, we can transform this equation into a statement in terms of probabilities,

$$\begin{aligned} (t+1)p_x(k,t+1) - tp_x(k,t) = \tilde{Z}[&\lambda_x(k-1)+\mu_x)p_x(k-1,t) \\ &- (\lambda_x(k)+\mu_x)p_x(k,t)] + \delta_{k,0} \end{aligned} \tag{22}$$

where we have simply used,

$$p_x(k,t) = \frac{n_x(k,t)}{N_x(t)} \tag{23}$$

Now, continuing to make use of the limit, $t \to \infty$, we can conjecture that the probability of seeing a node of type x with a given in-degree will settle to constant value:

$$\lim_{t\to\infty} p_x(k,t) = p_{x,\infty}(k) \tag{24}$$

this lets us rewrite (22) as,

$$\frac{p_{x,\infty}(k)}{p_{x,\infty}(k,t)} = \frac{k + \frac{\mu_x}{\lambda_x} - 1}{k + \frac{\mu_x}{\lambda_x} - \frac{1}{\tilde{Z}\lambda_x}} \tag{25}$$

where it is now useful to motivate,

$$\alpha_x = \frac{\mu_x}{\lambda_x} \tag{26}$$

and,

$$\gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x} \tag{27}$$

Now, if we restrict our domain to $k > 0$, we can use the known result that the form of (25) can be written in terms of the ratio of Gamma functions, giving us,

$$p_{x,\infty}(k) = \frac{A_x \Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)} \tag{28}$$

Therefore, if we can solve for the constant A_x, we can predict the probability of a given type of node having a given in-degree. The way solve for A_x is to consider (22) for $k = 0$. This gives us,

$$p_{\infty,x}(k = 0) = p_{0,x} = \frac{1}{1 + \tilde{Z}\mu_x} \tag{29}$$

and since (25) is an iterative equation valid when $k = 0$, we can use it to solve for $p_{x,\infty}(k = 1)$, which is within the domain where (30) is a valid. Matching the equation forms and solving for $A(k = 1)$, we find not only that,

$$A(k = 1) = p_{0,x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)} \tag{30}$$

but also the conjecture that,

$$A(k) = p_{0,x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)}, \forall k > 0 \tag{31}$$

Nicely, this is consistent to the extent that we expect this value to be a constant as A_x is a constant in the known result used to go from (27) to (30).

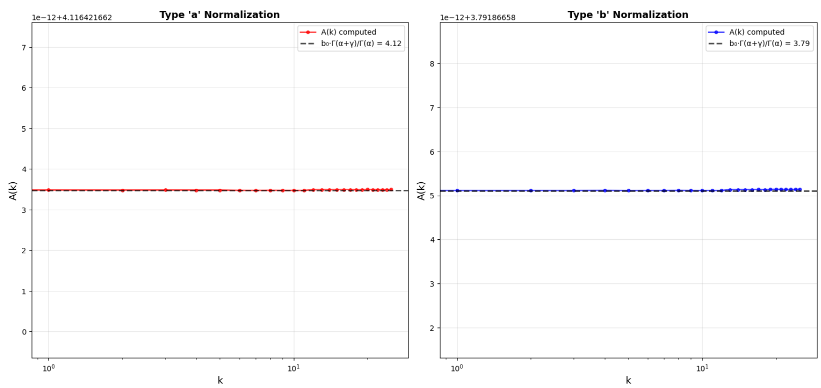

Figure 1: Numerical demonstration of the $A(k) \in \mathbb{R}^+$ conjecture for the integers $k = \{1, 2, \ldots, 25\}$. The network model is the simplified version where $x = a$ and $x' = b$, with $f_b = 1 - f_a$ and $h_b = 1 - h_a$. $f_a = 0.4, h_a = 0.8, \mu_a = 1, \mu_b = 2, N_{\text{initial}} = 100, N_{\text{total}} = N_{\text{initial}} + 25{,}000, m = 40$.

Substituting this A_x in, this gives us,

$$p_{x,\infty}(k) = p_0 \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)} \frac{\Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)} \tag{32}$$

Now using the Beta function, which can be defined as,

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)} \tag{33}$$

and reintroducing the $k = 0$ case, our equation can be written as,

$$p_{\infty,x}(k) = \begin{cases} p_{0,x} & k = 0 \\ p_{0,x} \cdot \dfrac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)} & k > 0 \end{cases} \tag{34}$$

Finally, by the symmetrical nature of this problem,

$$p_{\infty,x'}(k) = \begin{cases} p_{0,x'} & k = 0 \\ p_{0,x'} \cdot \dfrac{B(k, \alpha'_x + \gamma'_x)}{B(k, \alpha'_x)} & k > 0 \end{cases} \tag{35}$$

Therefore, aside from the $\langle g_x \rangle$ and $\langle g_{x'} \rangle$ terms, we now have an analytical solution for the in-degree distribution as $t \to \infty$, which depends on node type and the levels of homophily and heterophily in the system.

Furthermore, we can reduce these two unknown parameters to one, since, the total number of in-degrees in the network added at each timestep needs to equal the total number of out-degrees added at each timestep. In other words,

$$\langle g_x \rangle + \langle g_{x'} \rangle = m \tag{36}$$

Numerically, we see that not only do the proportion of in-edges per node type tend to constant values as $t \to \infty$, but these constants also sum to m.

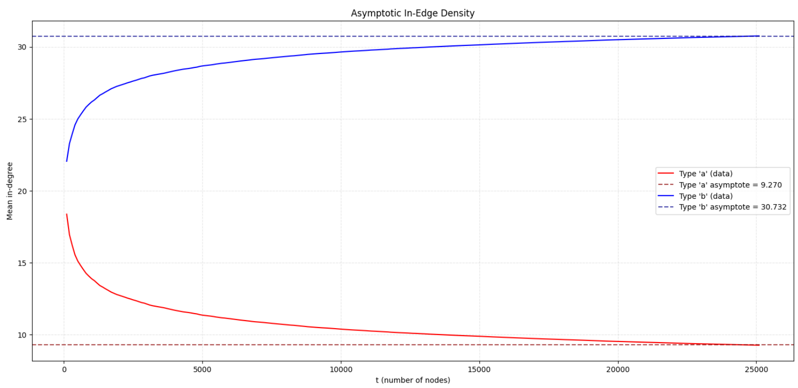

Figure 2: Numerical demonstration of the proportion of in-edges per node type tending to constant values as $t \to \infty$. This also shows these constants, as expected, sum to m. The network model is the simplified version where $x = a$ and $x' = b$, with $f_b = 1 - f_a$ and $h_b = 1 - h_a$. $f_a = 0.4, h_a = 0.8, \mu_a = 1, \mu_b = 2, N_{\text{initial}} = 100, N_{\text{total}} = N_{\text{initial}} + 25{,}000, m = 40$.

Numerical Model

Statistical Analysis of Numerical Model

The Cumulative Distribution Function for this is:

$$F_{\infty,x}(k) = \int_{-\infty}^{k} p_{\infty,x}(t)\, dt \tag{37}$$

But it appears the analytical solution to this is not a known result.

1 Misc

For simplicity, let

$$N(t_{\mathrm{initial}}) = t_{\mathrm{initial}} \tag{38}$$

and

$$E^{(\mathrm{in})}_{\mathrm{total}} = E^{(\mathrm{in})}_{x} + E^{(\mathrm{in})}_{x'} = f_x m\, t_{\mathrm{initial}} + f_{x'} m\, t_{\mathrm{initial}} \tag{39}$$

References

[1] Tim S. Evans. Personal communication. Dec. 2025.